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REVIEW



Deciphering the role of VapBC toxin-antitoxin systems in *Mycobacterium tuberculosis* stress adaptation

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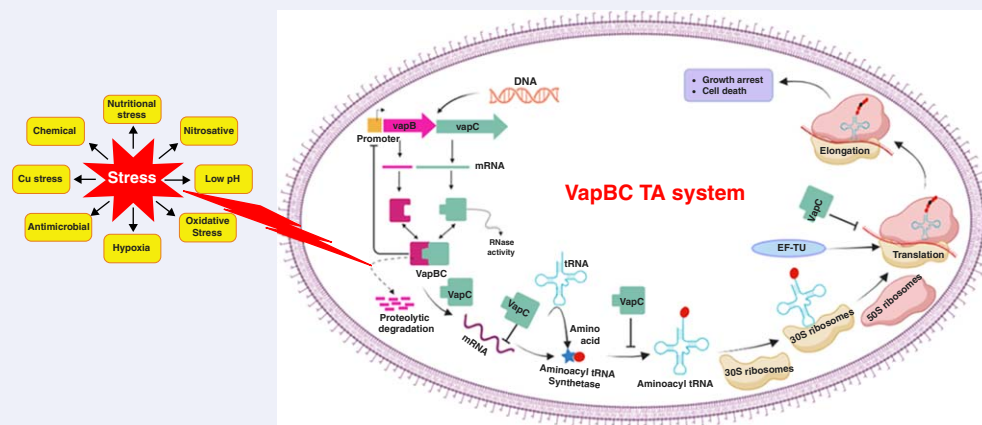
ABSTRACT

Mycobacterium tuberculosis (*Mtb*) harbors a high number of Toxin-Antitoxin (TA) systems, wherein half of them belong to virulence associated proteins B and C (VapBC) family that has a characteristic PilT N-terminus domain and ribonuclease activity. Functional insights into *Mtb* VapBC TA modules unraveled their role in adaptation to various host-mediated stressors, including oxidative/nitrosative, chemical and nutrient starvation as well as multidrug tolerance and establishment of persistence. To understand the intricacies of *Mtb*'s pathogenesis, absolute cellular targets of 19 VapC(s) were determined. Some exhibit a shared ribonuclease activity, whereas others harbor tRNAse and 23S rRNA cleavage activity. The detailed functional characterization of VapBC4, VapBC12 and VapBC22, including *in vivo* deletion mutant studies revealed their role in *Mtb*'s virulence/persistence. For example, the VapC22 mutant was attenuated for *Mtb*'s growth in mice and elicited a decreased T_H1 response, whereas mice infected with VapC12 mutant displayed a substantially higher bacillary load and pro-inflammatory response than the wild type, showing a hyper-virulent phenotype. Further experimental studies are needed to decode the functional role of VapBC systems and unravel their cellular targets. Taken together, *Mtb* VapBC TA systems seem to be promising drug targets owing to their key role in enduring stressors, antibiotic resistance and persistence.

PLAIN LANGUAGE SUMMARY

Mycobacterium tuberculosis (*Mtb*) is a type of bacterium that can make a person ill for a long time. This is because the bacteria can survive within the body, even in harsh conditions. Special genes called TA pairs help the bacteria to survive. *Mtb* has more than 80 of these pairs and the biggest group is called VapBC, which has 51 pairs. This review describes how VapBC TA systems help the bacteria to survive inside the human body. Since VapBC pairs help *Mtb* survive in our bodies, they could be good targets for new medicines.

GRAPHICAL ABSTRACT



Activation of *Mtb* VapBC TA system and its mode of action in response to different stresses are outlined.

ARTICLE HISTORY

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ATDs; *Mtb*; persistence; ribonuclease; toxin-antitoxin system; tRNAse; VapBC family; VapC toxin

1. Introduction

Tuberculosis (TB) caused by *Mtb* remains a significant global health threat, with 10.6 million new cases and 1.3 million deaths reported annually in 2022 [1]. The switching of active to latent TB assists in evading host-mediated stresses and antibiotic therapy, ensuring the long-term survival of this pathogen inside the host [2,3]. During *Mtb* infection, the bacilli come across several host stresses, in other words, nutrient starvation, hypoxia, acidic pH, oxidative stress and nitric oxide, which make transcriptional changes for their adaptation/survival, wherein the stressful conditions retard the bacterial metabolic activity and elicit a non-replicating persistent (NRP) state [4,5]. The majority of antibiotic-susceptible bacteria yield to host immunity and get killed, while a subpopulation of *Mtb* may transition into a persister state that survives high concentrations of antibiotics and host stressors without any genetic mutation [5,6]. Joshi et al. [7] meticulously described the molecular determinants involved in *Mtb*'s persistence and their therapeutic intervention, in which intrinsic mechanisms like transcriptional regulators, stringent response, lipid metabolism and cell wall remodeling enzymes are involved. Several *Mtb* genes are associated with bacterial persistence, namely, *icl1*, *relA*, *prcA*, *cydC*, etc. (involved in intermediary metabolism/respiration), *treS*, *hspX*, *acr2*, etc. (involved in virulence, detoxification and adaptation), *whiB3*, *dosR*, etc. (regulatory proteins), *pcaA*, *tgs1*, *hsaC*, etc. (involved in lipid metabolism) and *ripA*, *pstA1*, *phoY1*, *perM*, etc. (involved in cell wall and cell process) [7]. Additionally, the host macrophages, the main effector cells to control mycobacterial infection, produce reactive oxygen species (ROS) and reactive nitrogen species (RNS), where *Mtb* counteracts ROS/RNS stress by maintaining the cytosolic redox homeostasis and elicitation of ROS/RNS scavenger genes, i.e., *sodA*, *sodC*, *katG*, *mef*, *noxR1*, *noxR3*, *dlaT*, etc [8,9].

TA systems have also been implicated in growth regulation, drug tolerance and survival of *Mtb* in the NRP state [10,11]. Of note, TA systems are ubiquitous genetic modules comprising a stable toxin and labile antagonistic antitoxin that exist as tight complexes under normal growth conditions, but in response to stresses, i.e., nutrient starvation, antibiotic exposure, DNA damage, etc., the antitoxin partner is degraded and the toxin partner is liberated [12]. The unleashed toxin inhibits crucial cellular processes, viz. replication, transcription, translation, membrane integrity and cell wall biosynthesis, which ultimately leads to growth arrest and sometimes even cell death [13]. TA modules are regulated by differential stabilities of the highly expressed antitoxin and a stable toxin partner [12]. Antitoxin genes are usually located upstream of the toxin gene in the operon, except for the

HigBA TA [3,12]. Transcription of TA is most commonly autoregulated by binding of TA complex to its promoter region [12]. Though antitoxin partners can also bind to promoter regions, they have a lower affinity than the binding of the TA complex [3,12].

2. Types of TA systems

TA systems have been categorized into eight different classes based on their mechanism of action and the nature of interacting partners. In the Type I TA system, the antitoxin is a small, noncoding RNA (sRNA) that binds to and inhibits the translation of the toxin mRNA, preventing the production of toxic protein. Some Type I toxins can also disrupt cell membranes and ATP synthesis [14]. In the Type II TA system, both the toxin and antitoxin are notably proteins [15]. Under stress conditions, proteases degrade the antitoxin, releasing the toxin, which inhibits cell growth by targeting the essential cellular processes, i.e., DNA replication/translation, etc [15]. The Type III TA system makes a stable RNA-protein complex that prevents the toxin from functioning by degrading the RNA of the antitoxin through the action of an endoribonuclease [16]. In contrast, the Type IV TA system maintains the stability of bacterial filaments through protein antitoxin, yet the protein toxin causes destabilization [3]. In Type V TA systems, antitoxin functions as an RNase that specifically degrades the toxin mRNA, reducing its expression [14]. In the Type VI TA system, antitoxin acts as an adapter protein to direct the toxin to proteolysis, resulting in the toxin's inactivation [16]. However, if the toxin circumvents this degradation, it can still disrupt DNA replication by binding DNA to the sliding clamp [14]. In the Type VII TA system, antitoxins are proteins that deactivate their corresponding toxins through post-translational modifications [3]. In the Type VIII TA system, antitoxins are RNAs that suppress the expression of their corresponding RNA toxins, acting either as antisense RNAs or by recruiting a Cas9 protein to serve as a transcriptional repressor [14]. The main features of eight types of TA systems and their respective mechanisms for controlling TA module activation are outlined in Figure 1.

3. VapBC (virulence associated proteins B & C) systems of *Mtb*

Although *Mtb* TA systems have been reviewed for their roles in *Mtb*'s pathogenesis and stress adaptation [3,17–19], only two review articles have been published on the exclusive *Mtb* VapBC family [20,21]. Kang [20] provided the structural insights to crystal structures of six VapBC complexes of *Mtb* VapBC family, i.e. VapBC2, 5, 11, 15, 26 and 30, whereas Zarin et al. [21] described the cellular targets of *Mtb* VapC toxins and unpublished observations

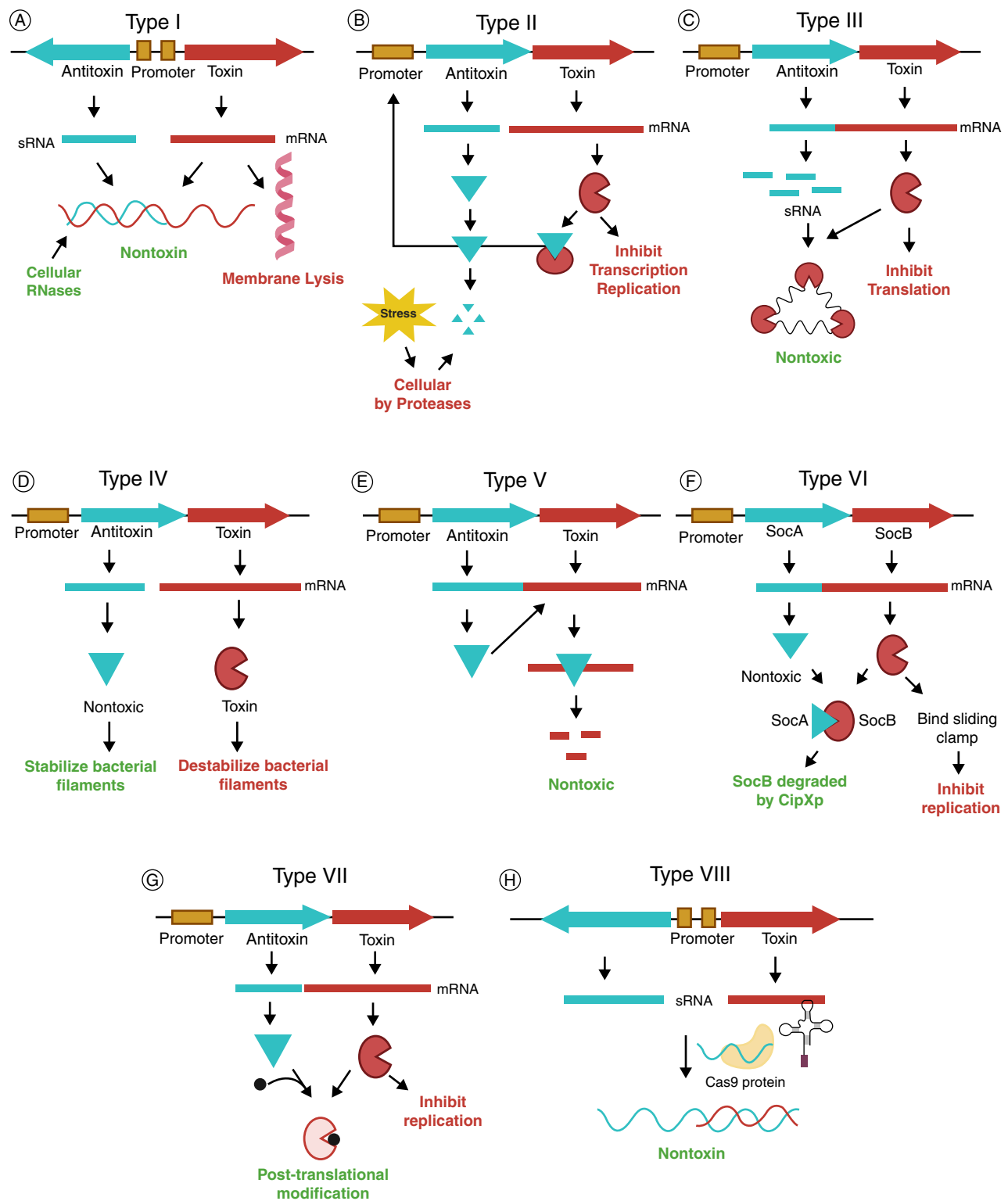


Figure 1. Key features of eight main types of toxin-antitoxin (TA) systems and their respective mechanisms for controlling TA module activation are outlined. Panels (A–H) schematically depict how the cognate antitoxins (turquoise color) neutralize the toxins (red color) in Type I–VIII TA modules; activities that are non-toxic are shown in green color, whereas those that are toxic are shown in red color. The genomic locations of TA modules are shown with colored arrows, while promoter positions are shown with yellow arrows.

on VapC16 toxin. In this review, we comprehensively evaluated the role of *Mtb* VapBC operons in stress adaptation, virulence and persistence as well as experimental findings on both individual/multiple VapBC systems till date. We also detailed the cellular targets of VapC toxins and their mechanisms of action, the same are presented in the tabulated form. Additionally, we critically analyzed the current methodologies utilized for functional investigations of VapBC operons, including the ectopic expression of VapC toxins and the use of transcriptional analysis as a metric for toxin activity. Markedly, *Mtb* harbors an expanded repertoire of TA systems, in other words, more than 80 altogether. Many TA loci belong to the type II category and are further grouped as VapBC, MazEF, ParDE, RelBE, HigBA, DarTG, FicAT, MbcAT and HipAB families [3]. Few of the *Mtb* TA members remain uncharacterized while being classified in the Type IV category. The VapBC family encompasses the largest number of TA systems in *Mtb*, i.e. 51 [3,22]. Despite reductive evolution, the abundance of VapBC members in *Mtb* highlights their significance in pathogenesis and intrigues further investigating their role in stress adaptation [3]. VapBC operon encodes a stable toxin (VapC) that contains a PiIT N-terminus (PIN) domain characterized by ribonuclease activity, $\alpha/\beta/\alpha$ sandwich topology and four conserved residues [22]. The VapC toxins exhibit a higher degree of similarity than their related antitoxins, which have undergone a significant divergence [23].

The initial investigation on *Mtb* VapBC TA operons was focused on establishing them as bona fide TA operons by evaluating their growth kinetics in a heterologous system or a native host [24–26]. Of note, 38 *Mtb* TA loci were overexpressed in *Escherichia coli* to determine their killing and rescue effect [24]. The ectopic expression of 10 toxins, i.e. VapC4, VapC11, VapC20, MazF3, MazF6, MazF7, MazF9, ParE1, ParE2 and HigB1 was found to be highly toxic for *E. coli*'s growth in liquid/solid media [24]. However, the co-expression of their respective antitoxin partners reversed the toxicity and established their identity as bona fide TA modules [24]. This investigation provided annotation to the previously unannotated TA gene sequences in the *Mtb* genome [24]. Further, to functionally characterize a higher number of *Mtb* TA loci (78), genome-wide comprehensive analyses were carried out which included VapBC family members to unravel their biological role in stress adaptation [25]. Furthermore, conditional expression of toxin and antitoxin proteins in *Mycobacterium smegmatis* resulted in the identification of 30 TA modules (among 78 analyzed) as bona fide operons, of which 19 belonged to the VapBC TA family. However, RNase activity of only VapC2 and VapC11 toxins was documented among the 19 bona fide VapBC operons [25]. Meanwhile, induction of Toxin Antitoxin Chaperone as well as VapC15 in response

to hypoxia and upregulation of VapC3 and VapC11 during macrophage infection suggest their involvement in stress adaptation [25]. Similarly, VapBC45 and VapBC49 were established as functional modules in heterologous systems, and the ribonuclease activity of toxinVapC45 was reported [27]. However, the detailed functional characterization of these operons has not yet been carried out.

4. Role of VapBC systems in stress adaptation

VapBC TA loci help in encountering various host-mediated stresses, such as hypoxia, oxidative, nitrosative, nutrient starvation, low pH and chemicals with their transcriptional reprogramming, which in turn enable *Mtb* bacilli to persist under harsh conditions [3,11,26]. To detect the key *Mtb* TA operons of cellular stress response and unravel a TA-co expression network, the expression profile of 79 *Mtb* loci including VapBC TA members was investigated by subjecting this pathogen to several stresses, including chemicals, nutrient starvation, low pH, etc [11]. Members of the VapBC TA family along with other TA family members were established as stress-responsive modules by demonstrating the transcriptional activation of each of the investigated TA operons after the pathogen's exposure to growth-limiting conditions [11]. The genome-wide study detected VapBC22, VapBC35, RelBE1, HigBA1 and HigBA3 to be upregulated in multiple stresses. Markedly, VapBC46 and HigBA2 were identified as the key loci that were activated in all the stresses being tested. Interestingly, Gupta et al. [11] constructed a biological network representing the induction and repression of investigated TA modules in response to various stresses, which represent TA gene regulation in the pathogen. Moreover, TA modules showing co-activation under various circumstances point out a shared regulatory mechanism [11].

Another meticulous study on the functional characterization of the *Mtb* VapBC TA system revealed that the ectopic expression of some VapC(s) inhibited the growth of *Mycobacterium bovis* BCG and altered the morphology of *M. smegmatis* [26]. The toxin members of the VapBC family were classified into three categories, i.e. highly active, moderately active, and inactive based on the growth kinetics of *M. bovis* BCG, wherein the toxin gene was placed under the control of anhydrotetracycline inducible promoter. Toxins such as VapC4, 11, 13, 20, 21, 22, 26, 27, 28, or 33 were classified as 'active toxins', since no significant increase in bacterial growth was seen in the overexpressed strains of *M. bovis* BCG, while the toxins VapC3, 15, 30, 34, 35, 36, 44, or 49 were classified as the moderately active toxins [26]. Further, to investigate the potential regulatory interplay among the VapBC operons and other TA families, mRNA expres-

sion profiling was carried out, wherein specific subsets of VapC toxins were induced in *Mtb* on exposure to oxidative/nitrosative, hypoxia and nutrient-limiting conditions. Notably, transcriptional cross-activation among the VapBC family members and other *Mtb* TA families has been reported [26]. The ectopic expression of VapC3, 11, 13, 20, 22, 27 and 33 resulted in significantly increased transcript levels of VapC15, while overexpression of VapC3, 11, 15, 20, 22 and 27 resulted in enhanced mRNA levels of HigB1. Moreover, MazF6 (Rv1991c) transcript levels were significantly upregulated in strains showing overexpression of VapC11, 13, 15, 20, 22 and 44. The simultaneous activation of these TA systems in various situations suggests a shared regulatory mechanism between the VapBC family toxins and other TA family toxins that allows *Mtb* to survive/persist inside the host. Additionally, the survival patterns of wild type and *Mtb* deletion mutants of VapBC3, VapBC4 and VapC28 were compared in response to hypoxia, nitrosative/oxidative stress, nutrient-limiting conditions and chemicals [26], which unraveled that individually VapBC3, VapBC4 and VapC28 were not necessary for *Mtb* bacilli to adjust against the analyzed stresses. Conversely, VapBC3 and VapBC4 have been shown to be essential for *in vivo* infection [26].

Several studies on *Mtb* VapBC TA systems utilized increased TA transcription as a metric for toxin activity [11,25,26]. However, increased TA transcription is not an accurate measure of toxin activity, since several investigations reported an increased TA transcription in response to stress but that does not correlate with the detectable toxin activity [28–30]. Of note, recent studies have documented that toxin activity is heterogeneous and not strong within a bacterial population, resulting in a mosaic of cells with varying metabolic rates [29,31]. Moreover, toxin activation is tightly controlled at various levels and strong toxin activity is unlikely in the bacterial population owing to several factors, *e.g.* high dissociation constant of TA complex, antitoxin abundance and requirement of specific triggers for toxin activation, among others [28,29]. In corroboration, some studies have shown that the techniques averaging across a bacterial population cannot capture the subtle toxin activity [29,31], which might mask the true heterogeneity; rather the single-cell approaches would be more appropriate.

5. Detailed functional studies on specific VapBC TA systems

Apart from the studies involving functional characterization of multiple *Mtb* loci, detailed functional investigations on the single VapBC loci such as VapBC4, 11, 12, 21

and 22 have been conducted to determine the role of concerned TA operons in *Mtb*'s pathogenesis against several stresses [10,32]. *Mtb* VapBC4, a typical TA operon, has been meticulously studied leading to its establishment as the functional operon since the ectopic expression of VapC4 resulted in growth arrest in heterologous systems and the native host [24,32,33]. The transcript level of *Mtb* VapC4 was upregulated in response to nutrient starvation, hypoxia and exposure to anti-tubercular drugs (ATDs), such as INH, levofloxacin and rifampicin (RIF), which underlines its role in stress adaptation, virulence and drug tolerance [11,26]. Moreover, Barth et al. [34] pinpointed the role of VapC4 in mediating the NRP state as well as enduring the copper and oxidative stress associated with *Mtb*'s virulence. Strikingly, deletion of VapC4 impaired the ability of *Mtb* to cause infection in animal systems, since VapC4 deletion mutant-infected guinea pigs exhibited a significantly decreased bacillary load (20-fold), as compared with wild type at both 4-week and 8-week post-infection [26].

Another TA system that has been meticulously characterized functionally and structurally is VapC11 [35]. The overexpressed VapC11 exhibited bacteriostasis in *M. bovis* BCG owing to its ribonuclease activity. Markedly, overexpression of VapC11 modulated the global transcriptional profile and overlapped with *Mtb* bacilli encountering hypoxia, nutrition starvation, or in drug-induced persistent state [35]. Intriguingly, overexpressed VapC11 caused the upregulation of 23 TA genes, proposing the occurrence of a greatly synchronized crosstalk between the *Mtb* TA systems [35]. These findings indicate that the unbound toxin could result in a significantly decreased *Mtb*'s metabolism, which is imperative for *Mtb*'s growth in macrophages [35].

Of note, phylogenetically similar VapBC2 and VapBC21 were also established as bona fide TA operons in heterologous and native host as per convention [36,37]. Functional characterization of VapC21 revealed that deletion mutant of VapC21 exhibited no significant effect on *in vitro* and *in vivo* growth of *Mtb*. However, transcriptome profiling of over-expressed VapC21 was like those observed in *Mtb* under stresses, such as nutrient starvation and hypoxia [36]. The overexpression of VapC21 has also led to an elevated expression of WhiB7 regulon and enhanced bacterial resistance to INH, RIF and levofloxacin, indicating the underlying significance of VapC21 in drug tolerance, thus proposing that there is an intricate web of connection between the non-matching TA pairs and VapC21 in drug tolerance [36].

Furthermore, VapBC22 has been extensively characterized by investigating the growth kinetics, transcriptome and proteome to find its contribution to *Mtb*'s physiology and pathogenesis to various stressors [38]. The

overexpressed VapC22 exhibited a bacteriostatic effect in the heterologous systems and native host, making the pathogen more susceptible to oxidative stress, but not to other stresses under laboratory conditions [38]. However, no effect on colony morphology, biofilm production, or persister formation was observed in the overexpressed strain [38]. The transcriptome profile of the overexpressed *Mtb* VapC22 strain exhibited augmented levels of ribosomal proteins and suppressed levels of metabolic enzymes, which is like *Mtb*'s response while enduring nutritional stress and hypoxia [38]. Conversely, proteomics analysis of the deletion mutant of VapC22 showed the reduced level of virulence proteins and an augmented level of cognate antitoxin proteins [38]. Markedly, the deletion mutant of VapC22 revealed an enhanced growth of *Mtb* in liquid media, although it was attenuated for *Mtb*'s growth in both guinea pigs/mice [38]. To decode the mechanism associated with attenuated phenotype, a transcriptome study was performed that unraveled the different gene expression in lung tissues of mice infected with VapC22 mutant/wild type [38]. In this study, ~145 genes were expressed differentially in VapC22 mutant vs. wild type at 4-week post-infection, while the transcripts of 97 and 48 genes were considerably enhanced and decreased, respectively, in the VapC22 mutant strain, when compared with the wild type [38]. Additionally, T_H1 response markers were evaluated from the lung homogenates of mice infected with wild type vs. VapC22 mutant, in which the interferon- γ (IFN- γ) transcript levels and lymphocyte antigen 6 gene expression were diminished, while the chemokines, i.e. CCL8, CCL17, etc. were considerably elicited in the VapC22 mutant, as compared with wild type [38]. In concordance with these findings, lung homogenates of mice infected with VapC22 mutant induced drastically lesser proinflammatory cytokines, including tumor-necrosis factor- α , interleukin-12 (IL-12) and IFN- γ but elevated anti-inflammatory cytokines, in other words, IL-4 and IL-10 than the wild type, suggesting a role of the innate immune system to clear the infection with VapC22 mutant strain [38].

Several functional characterization studies on *Mtb* VapBC systems used ectopic expression of VapC toxins to determine the TA dependent phenotype [26,35,36,38]. However, the ectopic expression of toxins doesn't ensure that the toxin within its native locus would induce the same phenotype. Unlike the overexpressed toxin, the toxin expression in its native state is heterogenous and subtle within a bacterial population [30]. Further, the ectopically expressed VapC toxins may not reflect the toxin's natural activity owing to TA stoichiometry and unique activation signals, thus the toxin must be studied in its native promoter system and response to

appropriate environmental cues [30]. Meanwhile, rather than determining the TA dependent phenotype through toxin overexpression, it would be preferable to detect the endoribonuclease activity of VapC toxins and perform well-designed deletion mutant studies under specific conditions to decode the TA-specific phenotype [30].

The major obstacle in current TB treatment is the NRP state and metabolically inactive subset of the *Mtb* population, the persisters [10,39]. Interestingly, the utility of host cholesterol as a carbon source has been shown for the maintenance of mycobacterial persistence during *Mtb* infection [40]. Talwar et al. [10] documented that cholesterol-mediated activation of ribonuclease toxin VapC12 has led to translation inhibition in *Mtb* by specific targeting of proT tRNA, which in turn augmented the frequency of persister generation in a heterogeneous *Mtb* population [10]. The deletion mutant of VapC12 failed to persist in a guinea pig model of *Mtb* infection, wherein the lungs and spleen of VapC12 mutant group had a substantially higher bacterial load, as compared with the wild type [10]. Subsequently, mice infected with VapC12 mutant elicited a pro-inflammatory response and showed a higher bacterial load (in lungs) and better ATD response than the wild type [39]. All the mice infected with wild type survived for 18 weeks in a high-dose experiment, while none of them survived in the VapC12 mutant group, authenticating the role of VapC12 in *Mtb*'s persistence [39]. Moreover, the VapC12 deletion evoked hyper-virulence in *Mtb* was reliant on TLR4 receptor, proposing the utility of VapC12 in coordinating the host innate immunity for the long-term survival of pathogen [39].

6. *in silico* studies on *Mtb* VapBC TA systems

In addition to experimental studies, our group computationally characterized the promoter regions and Protein-protein interaction network (PPIN) of genome-wide *Mtb* TA operons (79) including 50 VapBC family members [12,41]. The promoter regions of *Mtb* TA systems were examined by employing the knowledge-based and *de novo* motif discovery techniques, which documented the occurrence of σ^B regulatory motif in the promoter sequences of 43 *Mtb* TA modules, of which 28 belonged to the VapBC family [12]. Additionally, a unique conserved motif [(T/A)(G/T)NTA(G/C)(C/A)AT(C/A)] was recognized in the promoter sequences of 14 *Mtb* TA modules of which 12 belonged to the VapBC family, wherein antitoxin/TA complex can bind to regulate the transcription of TA operon negatively. Moreover, a computational study involving the construction and analysis of the PPIN of 79 *Mtb* TA system including 50 VapBC TA systems was carried out [41], which utilized PPI data extracted from the STRING v 10.5, yielding 486 PPIs. The bioinformatics study

identified the top 10 hub proteins as HigA1, VapB45, VapB14, Rv2762c, VapC1, VapC9, VapB11, VapC10, RelB and VapC19 by using the radiality and betweenness algorithm, wherein six out of the top 10 belonged to the VapBC family [41].

7. Cellular targets & mechanism of action of *Mtb* VapC toxins

Besides establishing the *Mtb* TA loci as bona fide operons, experimental work to identify the absolute cellular targets of 19 VapC(s) has been carried out, which described the strategies utilized by *Mtb* to survive/persist inside the host [2,33,36,37,42,43]. A comprehensive study to detect the cellular targets of 12 *Mtb* VapC toxins was performed by UV-cross linking, deep sequencing and phylogenetic analysis using *M. smegmatis* as a surrogate model [33]. Notably, all the investigated VapC toxins possessed endoribonuclease activity that significantly cut the RNAs for decoding at the ribosomal A-site [33]. The phylogenetically similar VapC toxins including VapC25, 29, 33, 37 and 39 exhibited cleavages against tRNA^{Trp-CCA} [33]. Of note, VapC15 and VapC32 toxins (which are phylogenetically like VapC11) showed cleavage activity against tRNA^{3-Leu-CAG}, while VapC11 targeted tRNA^{3-Leu-CAG}, tRNA^{13-Leu-GAG} and tRNA^{10-Gln-CTG}. Further, this study identified cellular target for VapC4, i.e. tRNA^{44-Cys-GCA}, although tRNA^{25-Ser-TGA} and tRNA^{28-Ser-CGA} were targets for VapC28 and VapC30, respectively [33].

Interestingly, VapC26 and VapC20 toxins demonstrated the cleavage of 23S rRNA in the Sarcin-Ricin Loop (SRL) [33]. These findings suggest that VapBC operons contribute to *Mtb*'s survival in human host by modulating bacterial growth in response to various stressors via inhibition of translation [33]. Another study involving functional characterization of five *Mtb* VapC toxins, i.e. VapC1, 19, 27, 29 and 39 was performed by employing pentaprobe RNA sequence and mass spectrometry, which revealed Mg²⁺ dependent RNase activity with UA*GG sequence as the optimal cut site for these toxins [42]. However, overexpression of these toxins displayed differential growth inhibition in *M. smegmatis* [42]. It is possible that *Mtb* VapBC TA systems having the same functions are regulated differently, leading to only a subset being activated in response to specific stresses. In addition to decoding cellular targets of multiple TA systems, the mode of action of a single VapC(s) has also been reported. For example, VapC4 has been shown to distinctly harbor sequence-specific endoribonuclease activity (ACGC and AC(A/U)GC) due to RNA binding and not RNA cleavage, as well as to display selective tRNA cleavage activity against tRNA^{Ser24}, tRNA^{Ala2} and tRNA^{Ser26} [43]. On the other hand, the recombinant

VapC11 cleaved 15 out of 45 *Mtb* tRNAs *in vitro* at a single location within their anticodon stem-loop to create tRNA halves. In contrast to broad tRNase activity exhibited *in vitro*, VapC11 expression in *Mtb* displayed a specific cleavage of only two tRNA isoacceptors, i.e. tRNA^{Gln32-CUG} and tRNA^{Leu3-CAG} that have the GG consensus sequence. Notably, *in vitro* studies are significant for identifying the class of RNA targeted and consensus sequences for the exact substrate recognition by endoribonuclease toxins [2], albeit the toxin needs to be expressed in the native host to absolutely detect the RNA target [2].

Additionally, the cellular targets of VapBC2 and VapBC21 were determined using *in vitro* assays, where both the toxins displayed partial or complete cleavage of several tRNAs and an efficient cleavage of tRNA^{fMet} [37]. Conversely, *in vivo* assays revealed that both the toxins solely inactivated the initiator tRNA^{fMet} at a single, identical position within their anticodon loop [37]. In agreement with the substantial role and essentiality of initiator tRNA^{fMet} in bacteria, the expression of each VapC toxin adequately impaired the translation, thus leading to growth arrest and subsequently cell death of *Mtb* [37]. Strikingly, Barth et al. [34] determined the cellular target and specific molecular processes set off by VapC4 that activated the stress survival pathways crucial for *Mtb*'s virulence. This study unraveled that VapC4 deactivated the only transfer RNA^{Cys} (tRNA^{Cys}) by cutting it at a single location within the anticodon sequence, though *in vivo* study demonstrated the cleavage of many *Mtb*'s tRNAs within their anticodon Stem Loop [34]. The exhaustion of tRNA^{Cys} repertoire resulted in the interruption of ribosome movement at Cys codons within mRNAs that were being actively translated [34]. Further genome mapping of these ribosomes stalled at the Cys position revealed several Cys containing open reading frames (ORFs) [34], which were not previously reported. Four of these short open reading frames (sORFs) encode the cysteine (Cys)-rich proteins (<50 amino acids) that serve as the Cys-responsive attenuators, causing ribosome pausing at Cys codon sequences to regulate the translation of genes downstream [34]. Of note, VapC4 imitates a Cys deprivation situation, which triggers the Cys attenuation at sORFs to universally orient the metabolism toward producing free Cys [34]. The Cys level contributes to mycothiol production, which is significant for the maintenance of cellular redox balance in *Mtb* during oxidative stress [34]. Additionally, the surplus Cys is utilized in specific cellular pathways, which assists the cells in defending themselves against the oxidative/copper stresses that *Mtb* typically encounters within macrophages [34].

Overall, *Mtb* VapC toxins mainly harbor the tRNase activity as observed in VapC2, 4, 11, 21, 25, 28, 29,

30, 33, 37 and 39. Contrarily, VapC1, VapC2, VapC19, VapC27 and VapC43 exhibit ribonuclease activity, while VapC20 and VapC26 display site-specific cleavage of the highly conserved SRL in 23S rRNA. A rich array of RNases with various regulation mechanisms may help in the cellular adaptation of *Mtb* bacilli to a wide range of situations, offering precise and dynamic control of gene expression to stressful stimuli. However, the cellular targets of majority of the VapC(s) are not yet known, more experimental work is required to decode the cellular targets so that the intricacies of *Mtb*'s pathogenesis can be unraveled. The cellular targets of *Mtb* VapC toxins and their precise mode of action are summarized in Table 1.

8. VapBC systems as drug targets

The establishment of VapBC operons in stress adaptation, virulence and drug resistance of *Mtb*, while the absence of their homologs in humans makes them highly attractive drug targets for ATD discovery [11,20,26,36]. VapBC TA system can be activated either by employing direct or indirect strategy for their utilization as drug target [44]. Direct activation of the VapBC TA system could be carried out either by preventing the formation of the VapBC TA complex or by inhibiting VapB translation [44]. Disruption of the VapBC complex by peptide-based inhibitors or small molecules could lead to the release of toxin from its antitoxin partner [44]. Novel antimicrobial peptides have been designed based on the crystal structure of VapBC30 that successfully disrupted the heterodimeric interface of the TA complex in a dose-dependent manner and triggered the RNase activity of VapB30 *in vitro* [45]. Similarly, the peptide-based inhibitors mimicking VapB11 were designed and tested in a dose-dependent manner against the VapC11 toxin [35]. Several toxin-mimicking peptides have also been developed by Kang et al. [46], which augmented the toxin activity by inhibiting VapBC26 complex formation, whereas blocking of antitoxin translation is another useful approach for toxin activation of TA complex. Antisense RNA can be used for this strategy since toxin and antitoxin genes are co-transcribed as a single mRNA but harbor separate Shine-Dalgarno sequences [44]. However, this approach has not yet been utilized for the activation of *Mtb* VapBC TA system.

Conversely, indirect activation of VapBC operons can be carried out either by enhancing the expression of protease degrading VapB or by triggering the VapBC operons via quorum sensing factors [44]. The combination of protease activating antibiotic acyldepsipeptide with RIF led to an absolute obliteration of *Staphylococcus aureus* biofilms *in vitro* as well as in a deep-seated mouse thigh infection model [47]. Similarly, this approach could

be applied to *Mtb* VapBC TA system for their usage as drug targets. On the contrary, extracellular death factor of *E. coli* and *Pseudomonas aeruginosa* (which acts as a quorum sensor) have been shown to artificially activate and enhance the endoribonucleolytic activities of MazF3/MazF6, but this indirect approach has not yet been tested in *Mtb* VapBC TA members [48]. Another promising approach is to inhibit the VapC toxin of a crucial *Mtb* VapBC complex so that VapC can be prevented from inhibiting the key cellular processes (e.g. translation) involved in modulating the growth, thus leading to the survival of bacteria in the NRP state. Moreover, the novel ATDs targeting VapC toxins have the potential to effectively combat drug resistant *Mtb*, which is a substantial challenge in TB chemotherapy [10]. The structural information of *Mtb* VapBC operons is the primary prerequisite to activate TA operons in the above studies for their utilization as drug targets [44,49]. However, the crystal structure of only seven *Mtb* VapBC members, i.e. VapBC5 (3DBO), VapBC11 (6A7V), VapBC15 (4CHG), VapBC20 (5WZ4), VapC21 (5SV2), VapBC26 (5X3T) and VapBC30 (4XGR) have been reported till date [49,50].

Unlike *Mtb* TA systems as the mediator of the persistence model, the focus has shifted away in *E. coli*, where the key persistence has been driven by the accidental prophage infection and many TA systems are involved in phage defense through mechanisms like abortive infection (Abi) [51,52]. Hames et al. [51] suggested the role of second messenger (p)ppGpp for *E. coli* persister formation via protease Lon and activation of TA modules. Owing to intrinsic toxicity, TA systems mediate Abi, where the host cell dies due to phage infection, eliminating the phage and protecting the clonal bacterial population [52]. However, neither all the TA systems are associated with phage defense, nor all the Abi systems are TA systems and there is an amalgamation between the toxin activity of TA systems and the outcome of initiated phage infection [52,53]. Although *Mtb* TA systems including VapBC are not shown to be phage defensive, the possibility of such a mechanism to elicit the persistence phenotype in *Mtb* cannot be ruled out.

9. Conclusion

VapBC is a type II TA system found in high numbers (51) in *Mtb*, which plays a substantial role in bacterial survival/persistence inside the host cells. Different subsets of VapBC TA systems are triggered in response to various stresses, i.e. oxidative/nitrosative, nutrient starvation, antibiotics, etc. Moreover, a regulatory interplay between them and the members of other *Mtb* TA families, MazEF/RelBE has been suggested. VapC toxins are typically endoribonucleases that target the mRNA,

Table 1. Different cellular targets of *Mtb* VapC toxins and their mode of action are summarized.

VapC toxin	Target	Mode of action	Ref.
VapC1	Optimal cut site of UA*GG	Sequence specific Mg ²⁺ dependent ribonucleases	[42]
VapC2	tRNA ^{fMet}	Initiator tRNA ^{fMet} inactivated by cleaving at a single, identical site within their anticodon loop, precisely between nucleotide 39 and 40	[37]
VapC4	1. tRNA ^{CysGCA} in <i>Mtb</i> 2. tRNA ^{Ala2} , tRNA ^{Ser26} and tRNA ^{Ser24} 3. tRNA ⁴⁴ ^{Cys-GCA} in <i>M. smegmatis</i> 4. Complete and partial cleavage of 8 and 19 <i>Mtb</i> tRNAs respectively (<i>in vitro</i>).	1. Essential prerequisite of the sequence U G ³⁴ C ³⁵ A ³⁶ A in the correct structural context for recognition and cleavage of tRNA ^{CysGCA} 2. Cleavage consensus sequences ACGC or ACUGC required within the anticodon loop 3. Not determined (ND) 4. ND	[33,34,43]
VapC11	1. 15 /45 <i>Mtb</i> tRNAs cleaved (<i>in vitro</i>) 2. tRNA ^{Gln32-CUG} and tRNA ^{Leu3-CAG} (<i>in vivo</i>) 3. tRNA ³ ^{Leu-CAG} , tRNA ¹³ ^{Leu-GAG} and tRNA ¹⁰ ^{Gln-CTG}	1. Cleavage at a single site within their anticodon stem loop to generate tRNA halves 2. Cleavages of two tRNA isoacceptors containing the GG consensus sequence 3. Cleavage of tRNA ³ ^{Leu-CAG} in the anticodon loop between nucleotide pairs G37↓U38, and U38↓G39 respectively	[2,33]
VapC12	proT tRNA	ND	[10]
VapC19	UA*GG sequences	Sequence specific Mg ²⁺ dependent ribonucleases	[42]
VapC20	23S rRNA in the SRL	Site specific cleavage of SRL in 23S rRNA	[33]
VapC21	tRNA ^{fMet}	Inactivated tRNA ^{fMet} by cleaving at a single, identical site within their anticodon loop precisely between nucleotide 39 and 40	[37]
VapC25	tRNA ⁷ ^{Trp-CCA}	ND	[33]
VapC26	23S rRNA in the SRL	Mg ²⁺ and Mn ²⁺ dependent ribonuclease	[33]
VapC27	UA*GG sequences	Sequence specific Mg ²⁺ dependent ribonuclease	[42]
VapC28	tRNA ²⁵ ^{Ser-TGA} and tRNA ²⁸ ^{Ser-CGA}	Cleavage site mapped to nucleotide pairs G36↓A37 in the anticodon loop	[33]
VapC29	1. UA*GG sequences 2. tRNA ⁷ ^{Trp-CCA}	Sequence specific Mg ²⁺ dependent ribonucleases	[33,42]
VapC30	1. tRNA ²⁵ ^{Ser-TGA} 2. tRNA ²⁸ ^{Ser-CGA}	Mg ²⁺ and Mn ²⁺ dependent ribonucleases	[33]
VapC33	tRNA-7 ^{TrpCCA}	ND	[33]
VapC37	tRNA ⁷ ^{Trp-CCA}	Cleavage at the anticodon loop at A36*A37*A38*A39, with the A37*A38 cleavage site most prominent.	[33]
VapC39	1. UA*GG sequences 2. tRNA ⁷ ^{Trp-CCA}	Sequence specific Mg ²⁺ dependent ribonucleases	[33,42]
VapC43	ND	Ribonuclease activity	[60]
VapC46	ND	Mg ²⁺ -dependent ribonucleases activity	[61]
VapC51	ND	Mg ²⁺ / Mn ²⁺ -dependent ribonuclease activity	[22]

Mtb VapC toxins showing weak activity or discrepancy in different systems are not included.

23S rRNA and tRNA molecules at specific sites. The *in vitro* studies unravel the targeted RNA class and the consensus sequences required for the endoribonuclease toxins, however, to accurately recognize the substrate, the toxin must be expressed in its natural host. Different studies on *Mtb* VapBC systems utilized transcriptional analysis to measure the toxin activity, though which is tightly regulated and depends on antitoxin abundance, TA complex stability, etc. Hence, TA transcription alone is not adequate to measure the toxin activity and future studies should also determine the endoribonuclease activity. Additionally, studies on *Mtb* VapBC(s) ectopically expressed the VapC toxins to determine the TA-specific phenotype that may not reflect the toxin's natural activity owing to TA stoichiometry and unique activation signals. Hence, the toxin must be monitored in its native promoter system and response to pertinent environmental cues. Nonetheless, VapBC4, VapBC12 and VapBC21 turn out to be promising in stress adaptation, persistence and drug tolerance, respectively, thus they can be harnessed as prospective drug targets. To date, the meticulous

functional characterization of VapBC4, 11, 12, 21 and 22 have been carried out and cellular targets of only 19 VapC(s) with no discrepancy have been determined. More research is needed to elucidate the role of the remaining 46 *Mtb* VapBC(s), cellular targets of 32 VapC(s) and crystal structure of many VapBC members to unravel the cellular and molecular mechanisms associated with *Mtb*'s persistence.

10. Future perspective

During recent years, the *Mtb* VapBC TA systems have drawn considerable attention due to their role in stress adaptation, persistence and antibiotic tolerance. However, functional characterization including the identification of absolute cellular targets of most of these operons has not been carried out. Further research would investigate the environmental cues that trigger their activation and detect the cellular targets, which may provide insight into how these systems contribute to bacterial adaptation/survival inside the host. In fact,

TB treatment has become quite challenging owing to antibiotic tolerance exhibited by the *Mtb* persister population against the traditional ATDs [10]. Hence, there is an urgency to design new drugs that may particularly kill the persister population. Given their importance in adaptation, bacterial persistence and antibiotic tolerance, the *Mtb* VapBC systems represent attractive targets for drug discovery [10]. The VapBC TA system can be induced via direct or indirect approach for their utilization as drug targets [44]. Direct activation involves stalling the formation of the VapBC complex or inhibiting the antitoxin translation. Alternatively, activation of VapBC operons occurs indirectly either by increasing the expression of a protease that degrades the VapB or stimulating VapBC operons using the quorum sensing factors [44].

Among the *Mtb* VapBC family, VapC4, VapC12 and VapC21 toxins seem to be promising drug targets as their role in persistence/drug tolerance has been well documented [10,34,36]. By targeting these toxins using the computer-aided drug design strategy including virtual ligand screening (VLS), the Chembridge compound library would be screened against the druggable pocket of toxin protein [54,55]. The top scorer compounds obtained via VLS would be further tested by RNase assay and evaluating their activity on whole *Mtb* bacilli *in vitro* [56], which would identify the potent *Mtb* VapC inhibitors. Subsequently, after evaluating the effect of those inhibitors in non-replicating/persistent *Mtb*-infected RAW264.7 murine/THP-1 human macrophages [5,57], the short-listed lead compounds would be subjected to pharmacokinetics/pharmacodynamics analyses in mice [58,59]. Future investigation of VapBC TA systems may unravel the precise mechanisms associated with persistence and identify more drug target candidates to deliberate the efficient therapeutics.

Article highlights

- *Mtb*, the causative agent of tuberculosis, utilizes TA systems to regulate stress response, persistence and antibiotic tolerance.
- *Mtb* possesses more than 80 TA systems.

Types of TA systems

- TA systems are categorized into eight distinct types according to their mode of action and the characteristics of their interacting partners, wherein type II is the most investigated class.

VapBC (virulence associated proteins B & C) systems of *Mtb*

- VapBC is the largest TA family (51) of *Mtb* that encodes a stable VapC toxin and unstable VapB antitoxin partner.
- Initial studies identified and validated several *Mtb* TA systems, including VapBC, as bona fide operons.

Role of VapBC systems in stress adaptation

- VapBC TA operons in *Mtb* are associated with several stresses (oxidative/nitrosative, copper, hypoxia, nutrient starvation, chemical) and are involved in *Mtb*'s survival in the non-replicating persistent state.

- Co-transcription of *Mtb* VapBC members to several stressors indicates a shared regulatory mechanism in response to cellular insults encountered by the pathogen.

Detailed functional studies on specific VapBC TA systems

- *Mtb* VapBC4, 11, 12, 21 and 22 play crucial roles in stress adaptation, virulence and drug tolerance, whereas VapC4 and VapC12 are particularly significant in *Mtb*'s persistence.
- Functional studies reveal that overexpression of VapC11 and VapC22 modulate the transcriptional profiles and enhance resistance to stresses, indicating a complex network vital for *Mtb*'s survival and drug tolerance.

Cellular targets & mechanism of action of *Mtb* VapC toxins

- *Mtb* VapC toxins are majorly RNase that exhibit the site-specific cleavage against mRNA, tRNA and 23S rRNA.

VapBC systems as drug targets

- *Mtb* VapBC TA systems seem to be promising drug targets owing to their ability to induce bacterial killing and their potential to overcome drug resistance.

Competing interests disclosure

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

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